
Analyzing Factors that Cause Alzheimer Disease

Kaiming Chen
KC1307

Ying Xiong
YX365

1 Introduction

What early-life and mid-life factors are most strongly associated with Alzheimer's disease risk in older adults?

We plan to solve the above question by analyzing two different datasets that contain information on lifestyle, medical history, and demographic that might cause Alzheimer's disease. The primary focus is going to be on the one of the dataset that has more structured features relevant to the study. Analysis through visualizations will be done on the dataset to check the relevance of the features. A supervised model will be trained using the dataset to solve the question proposed.

This approach is related to what we learned in class that we will use pandas data frame to read and store the datasets and analyze them through visualizations using matplotlib/seaborn. A supervised learning model is selected and trained using the dataset to evaluate the relevance of these factors, in order to see what kind of features are most strongly related with the disease. At the end, the model's performance will be measured by calculating the precision and accuracy which is also taught in class.

1.1 Context and Motivation

The research question is interesting because of how Alzheimer is related to memory loss, which memory is an important part of humans and it affects how human behave. The first thought that came across was about how sleep affects memory due to various reasons including stress, course work and jobs cause people to have sleep deprivation. There are a lot of people say that sleep would affect one's memory, which it makes one to question whether or not it is true. A more specific and related research topic in the health field is about Alzheimer disease, which at certain age, especially for people who are older have significant memory loss and they start to behave like children. It is hard to measure memory loss, while with the measure of Alzheimer disease, one could perform some analysis regarding the datasets about Alzheimer diseases and some factors that affect people to have Alzheimer disease.

1.2 Background Knowledge

Alzheimer's disease (AD) is a neurodegenerative disease that usually starts slowly and progressively worsens. It is the cause of 60–70 Percent of cases of dementia. The most common early symptom is difficulty in remembering recent events. As the disease advances, symptoms can include problems with language, disorientation (including easily getting lost), mood swings, loss of motivation, self-neglect, and behavioral issues. As a person's condition declines, they often withdraw from family and society. Gradually, bodily functions are lost, ultimately leading to death. Although the speed of progression can vary, the average life expectancy following diagnosis is three to twelve years.

Some of existing questions in this area include:

1.3 Known Risk Factors(Correlation and Causation)

Lifestyle and Behavioral Factors

- **Sleep Quality:** Poor sleep reduces the brain's glymphatic clearance of amyloid beta, a protein implicated in Alzheimer's pathology.
- **Diet Quality:** Adherence to Mediterranean and DASH diets, rich in fruits, vegetables, and healthy fats, is associated with lower AD risk.
- **Physical Activity:** Aerobic activity supports cognitive function and may delay neurodegeneration.
- **Alcohol Consumption:** Excessive alcohol intake may impair brain function, although the relationship with AD remains complex.
- **Smoking:** Smoking may cause brain tissue related damage and long-term smoking may lead to cognitive impairment indirectly affecting Alzheimer's disease..

Clinical and Metabolic Factors

- **Body Mass Index (BMI):** Obesity and underweight have both been associated with increased dementia risk, especially in late life.
- **Diabetes:** Type 2 diabetes is a known metabolic contributor to neurodegeneration and vascular dementia.
- **Hypertension:** High blood pressure contributes to vascular injury and cognitive decline.
- **Depression:** Chronic depression may lower cognitive reserve and precede Alzheimer's onset.
- **Stress:** Sustained stress may impair hippocampal function, increasing the risk of cognitive decline and the likelihood of other neurologically related changes in the brain..

Sources: 'A 2 year multidomain intervention of diet, exercise, cognitive training, and vascular risk monitoring versus control to prevent cognitive decline in at-risk elderly people (FINGER): a randomised controlled trial'

2 Method

2.1 Identifying Data Source

We used the publicly available dataset from Kaggle as the basis for our study, to avoid the subjectivity and limitations of a single data subject. We used two datasets to cross-validate each other. Both of the datasets are in the form of comma separated value (csv) file. The main dataset, Alzheimer's Disease Dataset, contains anonymous health data on 2,149 individuals aged 60 years and older. These individuals are labeled as either Alzheimer's Disease (AD) patients or normal healthy individuals.

The main dataset, Alzheimer's Disease Dataset (df1¹), includes 35 columns covering demographic variables (Age, Gender, Ethnicity, Education Level), lifestyle factors (BMI, Smoking, Alcohol Consumption, Physical Activity, Diet Quality, Sleep Quality), medical history (e.g., Diabetes, Hypertension), clinical metrics (Blood Pressure, Cholesterol), and cognitive and functional scores (MMSE, Functional Assessment, ADL). ADL represents the Activities of Daily Living score, and MMSE represents the Mini-Mental State Examination score; for both, lower scores indicate greater cognitive impairment. Most health and symptom indicators are binary, while lifestyle and assessment scores are on standardized scales. The dataset contains all numerical values—binary or ordinal—except for the DoctorInCharge column.

The second data set "Alzheimer's Prediction Dataset(df1²)", contains 74,283 individuals across the world who are also labeled as either diagnosed with Alzheimer's disease or not. The second dataset contains 25 columns that are features such as demographic variables (Age, Gender, Employment Status, Education Level), lifestyle factors (BMI, Physical Activity Level, Smoking Status, Alcohol Consumption, Sleep Quality, Dietary Habits), medical history (Diabetes, Hypertension, Cholesterol Level, Family History of Alzheimer's, Stress Level,) and cognitive score (Cognitive Test Score). Most of the features are similar except for some features that did not appear in the main dataset

¹<https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset>

²<https://www.kaggle.com/datasets/ankushpanday1/alzheimers-prediction-dataset-global>

including(Air Pollution Exposure, Genetic Risk Factor, Social Engagement Level, Urban vs Rural Living). The second dataset contains most of the columns labeled with phrases such as "low", "medium", and "high", some columns like Age and BMI are labeled with numerical values.

Prior to analysis, the dataset was preprocessed to normalize continuous variables and correctly code categorical features. Readability for later plotting and organizing was facilitated.

These two datasets are chosen because they contain features that are possibly factors related to Alzheimer's disease and also provide the data of whether these people are diagnosed with Alzheimer's or not. The data is ideal to answer the research question when used to train the supervised model to find the relationship.

2.2 Define Problem/Feature Space

The research question surrounds the binary classification of Alzheimer's diagnosis, with 1 = diagnosed, 0 = not diagnosed. The columns represent potential predictors and grouped based on domain knowledge(demographic, lifestyle, medical, cognitive). The features are selected and encoded based on relevance and data type (binary, ordinal, numeric).

2.3 Data Analysis

The analysis began with macroscopic statistics to help explore the demographic composition as well as to explore if there were patterns. Gender and race distributions are shown using Figure 1. This bar chart shows that Alzheimer's diagnosis rates vary by age group and gender. Overall, there is no significant increase of rates as age increases, while males exhibit slightly higher rates than females in older age ranges.

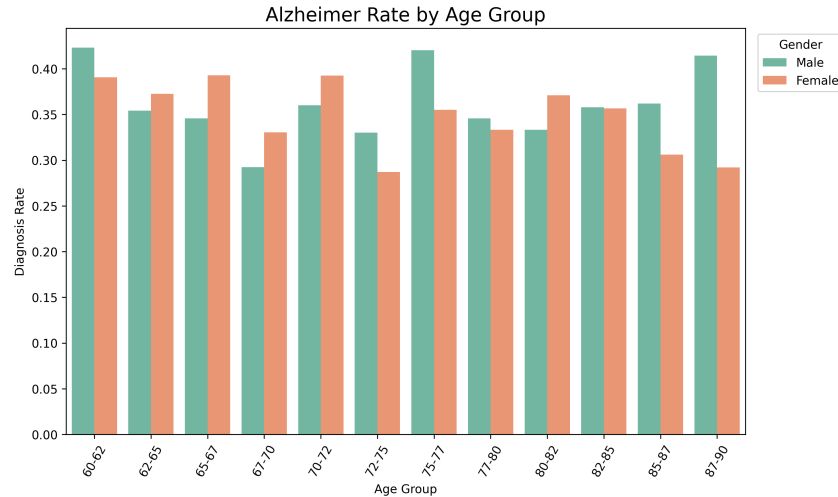


Figure 1: Barplot of Alzheimer Rate by Age Group using df1

The heatmap in Figure 2 shows the highest correlation with Alzheimer's diagnosis being the Functional Assessment feature with -0.36, which is not highly correlated, however other features are having even lower correlation that are less than 0.1. The heatmap in Figure 2 suggests that while no single feature linearly predicts Alzheimer's diagnosis, several clusters of features (particularly cognitive and functional assessments) may jointly contribute to the diagnosis of disease, supporting the use of non-linear models for classification.

The scatterplot in Figure 3 compares alcohol consumption with functional assessment scores, colored by diagnosis status. The blue and red dots are spread across x-axis indicating no linear relationship is found between Alzheimer's diagnosis and alcohol consumption. While there are significantly more clusters of blue dots (Alzheimer's patients) at low functional scores (0-4), and healthy individuals are more prevalent at higher functional scores (6-10). There is no strong correlation between alcohol consumption and functional status.

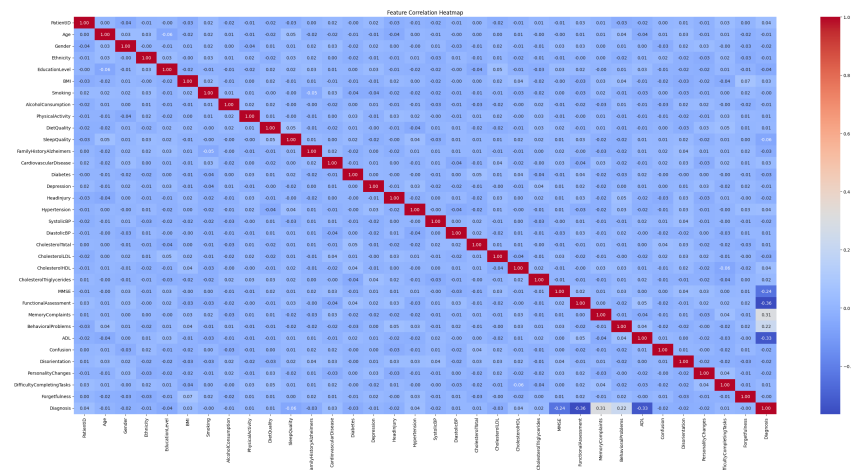


Figure 2: Heatmap of Feature Correlations in df1

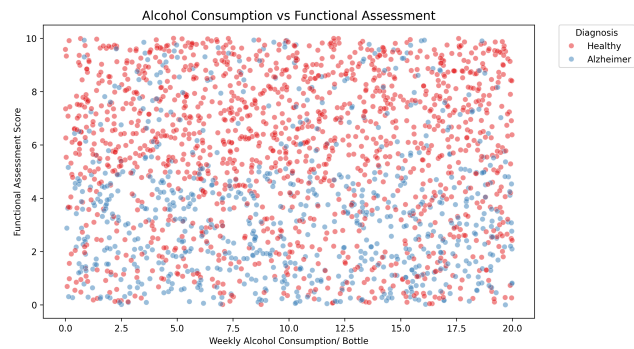


Figure 3: Scatterplot of Alcohol Consumption vs Functional Assessment by Diagnosis

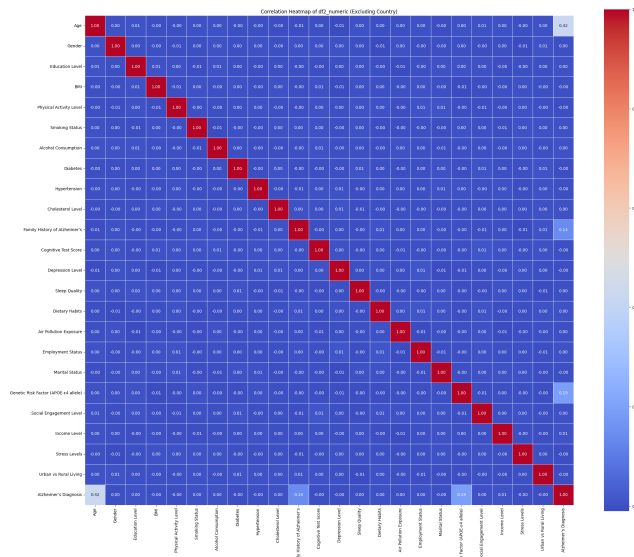


Figure 4: Heatmap of Feature Correlations in df2

The heatmap in Figure 4 is showing the correlation of the features for the second dataset, df2. It Supports the data in df1 that there is no single feature that has a strong correlation with the diagnosis

of Alzheimer's. However, this dataset is not the primary focus because most of the feature are labeled with 0.00 which is not useful to extract any patterns.

Other than the four figures presented here, there were more graphs including boxplot, density curve and bar chart to compare the features against each other to see any correlation. All the plots support that there is no single feature that shows a linear relationship with the diagnosis of Alzheimer's, thus a cluster of features should be examined together by the model.

2.4 Implementation steps

The entire analysis was implemented in **Python**, using **Pandas** and **Numpy** for data processing, **Matplotlib** and **Seaborn** and **Plotly** for graph visualization, and **Scikit-learn** for modeling and dimensionality reduction and noise reduction.

- **Data Cleaning and Transformation:** Replace categories such as gender and race with interpretable labels and saved it as a labeled data frame. For example, gender was converted to male according to 0 and female according to 1. All numeric columns have been replaced with definitions that are easier to draw and easier to read.
- **Exploratory Visualization:** To explore the relationship between Alzheimer's disease and the variables in the database, we created over 12 graphs. These included bar charts for gender and race for macro-analysis of the diagnosed population. There were also bar charts based on diagnosis rates by age and sex and correlation heatmaps for all variables at the macro level. correlation heatmaps and binary conditional plots were used to explore whether there were associations between the variables.
- **Modeling:** Random Forest was chosen as the supervised learning model because, according to the analysis, there are complex interactions between risk factors. Random forests are capable of modeling interactions between nonlinear decision features, whereas logistic regression can only model linear boundaries. We trained a random forest classifier to compare features and predict Alzheimer's diagnosis. Variables were scaled using models such as StandardScaler. Model performance was evaluated using metrics such as accuracy and ROC-AUC. Feature values were extracted to help evaluate each variable.

3 Results

3.1 Performance of Model

We trained a Random Forest classifier with 100 estimators to predict whether the individual has Alzheimer's or not using a mix of lifestyle, demographic, and clinical features. The first data set, df1, was split into 80% training and 20% testing sets. A random state of 42 is chosen, while any integer will work, the result might be different by a little.

The model demonstrated strong predictive performance, achieving an overall accuracy of 93%, with a ROC-AUC score of 0.95. Based on the classification report, the recall for the patient without Alzheimer is the highest, 0.97, which means that there are few false negatives for the individuals without Alzheimer. The lowest score is 0.86 which is the recall for individual with Alzheimer's. There are some people who are not diagnosed with Alzheimer by this model. Precision scores are the same for people with and without Alzheimer's, meaning that false positives do not occur frequently. Most of the individuals are predicted correctly by the Random Forest model.

3.2 Comparison to other methods

Random Forest was chosen for its ability to capture complex, nonlinear relationships between features and the diagnosis outcome. Compared to simpler models like logistic regression, Random Forest achieved substantially better classification metrics, including a ROC-AUC of 0.95. A logistic regression model is also trained as a comparison, while the ROC-AUC score is 0.89 with accuracy score of 0.83.

3.3 Visualizations

A horizontal bar plot is used to showcase the result from the model organizing the features based on their relevance score. The X-axis has the feature importance scores (how much each feature contributed to the model's predictions). The Y-axis contains the feature names (eg. FunctionalAssessment, ADL, MMSE). The bar plot provides a clear ranking of the feature influence and visually compare many features side by side.

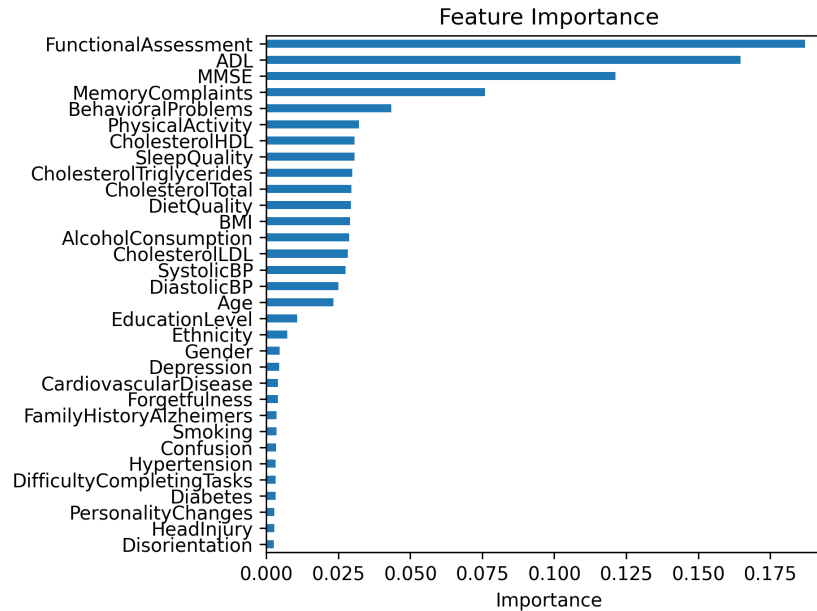


Figure 5: Feature Importance Ranking by Random Forest Model

4 Discussion

4.1 Anticipated Findings

Prior to conducting the analysis, we hypothesized that Alzheimer's diagnosis would be most strongly associated with cognitive decline due to poor sleep quality, low physical activity, and high BMI. These expectations were based on the public health research we have done linking cognitive outcomes to behavioral factors.

We also anticipated that medical issues such as diabetes, hypertension, and depression would be positively correlated with Alzheimer's risk. We thought that poor sleep and diet quality plus the alcohol consumption rate would have a correlation with cognitive decline, including a low functional assessment score, which further causes the diagnoses of Alzheimer's disease. Additionally, we predicted that cognitive (MMSE) and functional assessments would be substantially lower in the diagnosed group, reinforcing diagnosis validity.

4.2 Actual results

Based on Figure 5 created by the Random Forest model that was trained using the dataset, df1, it shows the three features that have the highest correlation are functional assessment, ADL and MMSE. These three features, however are not life factors, they are assessment on cognitive skills. The most correlated life factors according to Figure 5 are physical activity, sleep quality and diet quality respectively in the order. The life factors are less significant compared to the first three factors, which only relying on the life factors might not correctly predict the diagnosis of the individuals. However they are still included in the top 11 important feature that are used to predict the diagnosis of Alzheimer's.

Most of the medical factors remain low in the ranking of Figure 5, indicating the insignificance of these factors like diabetes, cardiovascular disease and hypertension. They contribute to less than 0.05 importance to the model, which without them the result of predicting the diagnosis of individual would not change much.

5 Conclusion

In summary, the aim of this report is to explore the main links between Alzheimer's disease diagnosis and lifestyle. By using a dataset containing 2,149 individuals and supporting the analysis with an auxiliary dataset to enhance rigor. We applied visualization and Random Forest modeling techniques to explore patterns and summarize. Both our visualization results and Random Forest modeling results suggest that PhysicalActivity, SleepQuality, and DietQuality are the lifestyle factor variables most strongly associated with an Alzheimer's disease diagnosis. These patterns are the same as we initially expected. Lifestyle variables such as diet, physical activity, and sleep quality do have a significant impact on physical condition.

6 Future Work

Although we believe that this report provides important insights into the influencing factors associated with Alzheimer's disease and that we used multiple database cross-checks for reporting fairness. However, this data still has limitations. Future research requires more effort

- Expand the database as well as the scope to include populations from different regions of the globe as well as ethnicity to stabilize the model.
- Cross-sectional future treatment data to track disease progression over time and make relative expectations.
- Intersect with other disciplines and join members of biochemistry fields such as genomics and neuro imaging for deep data processing research.
- Use high breadth and depth of learning data to specialize advanced models for accurate prediction
- Conducting a philanthropic network analysis to help physicians quickly conduct early screening efforts

7 Reference

A 2 year multidomain intervention of diet, exercise, cognitive training, and vascular risk monitoring versus control to prevent cognitive decline in at-risk elderly people (FINGER): a randomised controlled trial Ngandu, Tiia et al. The Lancet, Volume 385, Issue 9984, 2255 - 2263 [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(15\)60461-5/abstract](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(15)60461-5/abstract)

Lupien, S., McEwen, B., Gunnar, M. et al. Effects of stress throughout the lifespan on the brain, behaviour and cognition. Nat Rev Neurosci 10, 434–445 (2009). <https://doi.org/10.1038/nrn2639>

Wang, J., Chen, C., Zhou, J. et al. Integrated healthy lifestyle even in late-life mitigates cognitive decline risk across varied genetic susceptibility. Nat Commun 16, 539 (2025). <https://doi.org/10.1038/s41467-024-55763-0>

Kharoua, Rabie El. "Alzheimer's Disease Dataset." Kaggle, 11 June 2024, www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset.

Panday, Ankush. "Alzheimer's Prediction Dataset (Global)." Kaggle, 30 Jan. 2025, www.kaggle.com/datasets/ankushpanday1/alzheimers-prediction-dataset-global.